

WHAT LIES BEYOND THE CUT

How much turbulence can a coarse grid know about?

Approaching the optimal closure: equivariance, inductive bias, and Reynolds-number generalization in data-driven LES

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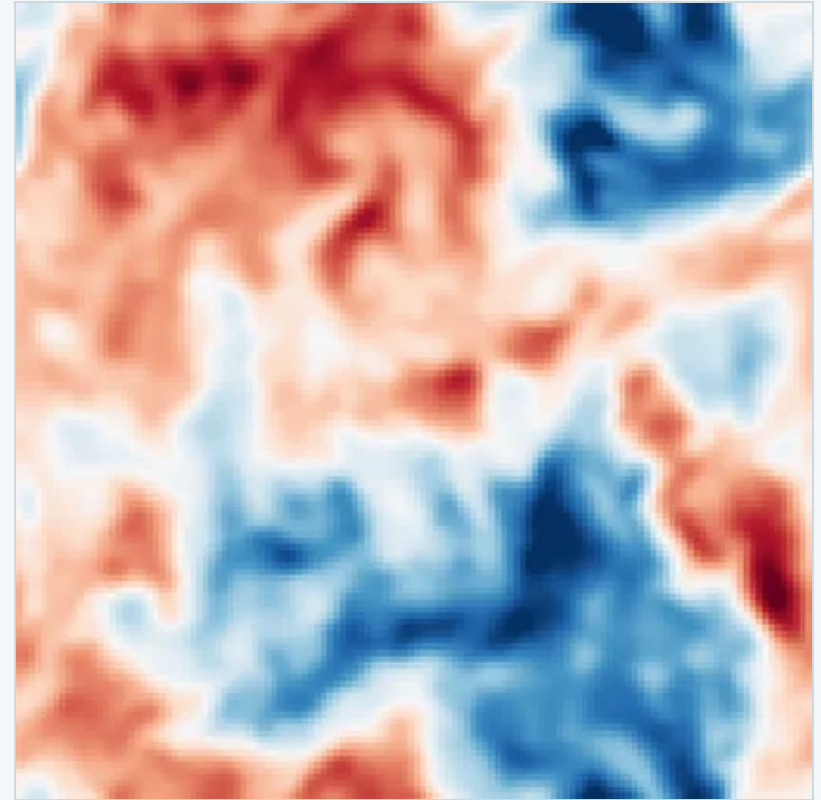
Centrum Wiskunde & Informatica, Amsterdam · Eindhoven University of Technology

WCCM-ECCOMAS 2026 · Munich · 22 July

MS184A · Theory-guided Design of Deep Learning-based Surrogates

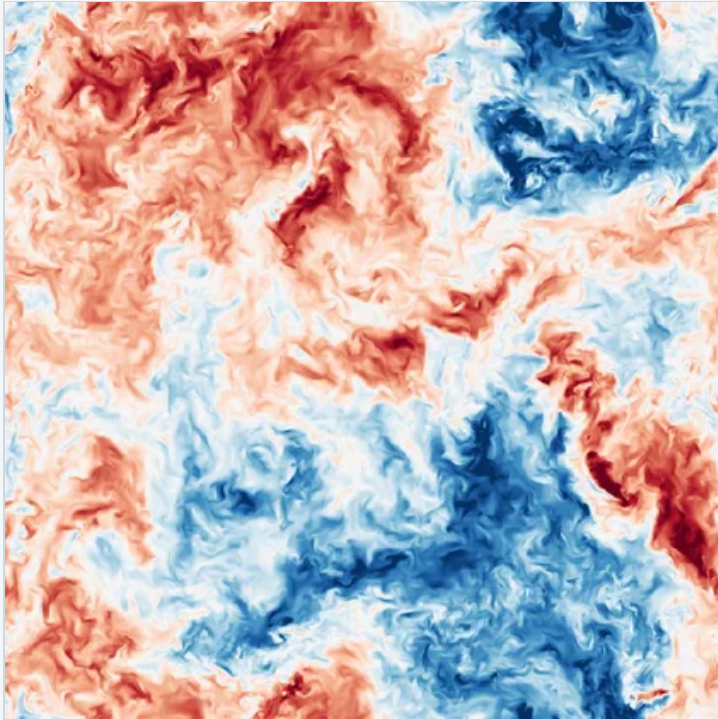
Funded by NWO Talent Programme Vidi (VI.Vidi.193.105)

compute on SURF / Snellius (EINF-15798)

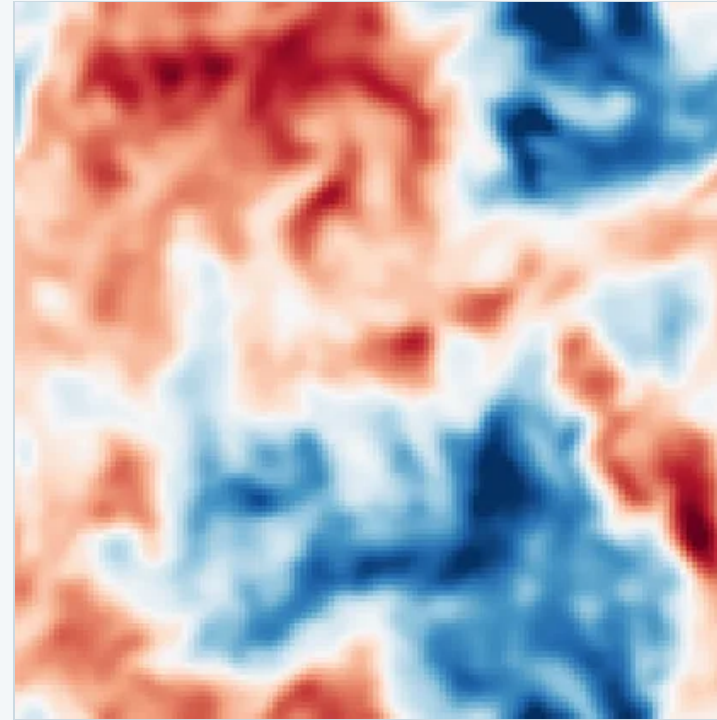


what an LES grid sees of a turbulent flow

The same flow, on two grids



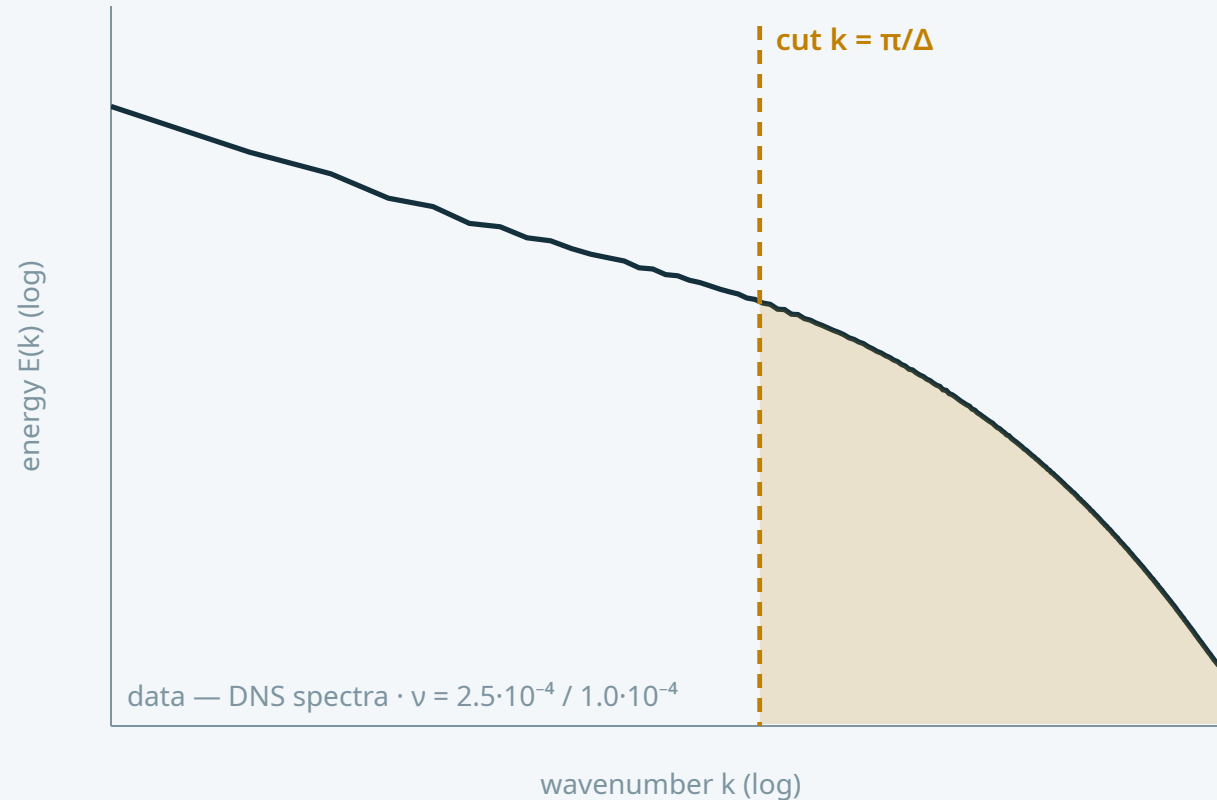
DNS · VELOCITY u
 810^3 — resolves everything



LES · FILTERED VELOCITY $\bar{u}$
 128^3 — 250× fewer points, misses the small scales

the missing scales still act on the resolved ones — that action is the closure's job

Where the energy lives — and where the grid stops



the grid resolves everything left of the cut;
the shaded energy is invisible

coarsen the filter ($\Delta \uparrow$) — more energy hidden

lower the viscosity ($\nu \downarrow$) — the spectrum grows a longer tail

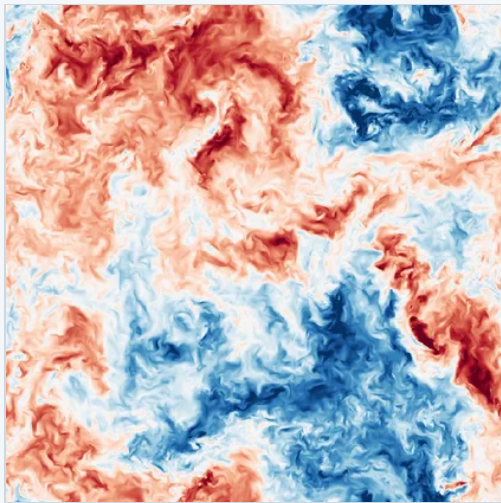
$$\text{Re}_\Delta = \frac{\Delta^2 \|\bar{A}\|}{\nu}$$

one resolved-scale number for

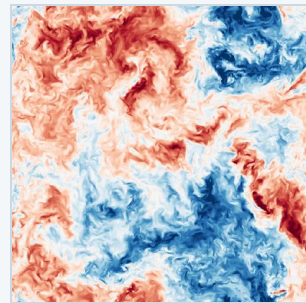
how much lies beyond the cut

$\bar{A} = \nabla \bar{u}$ — the resolved velocity gradient · $\|\bar{A}\|$ its r.m.s. over the box

Why this number? Scale the equations and see what survives



1
a turbulent solution



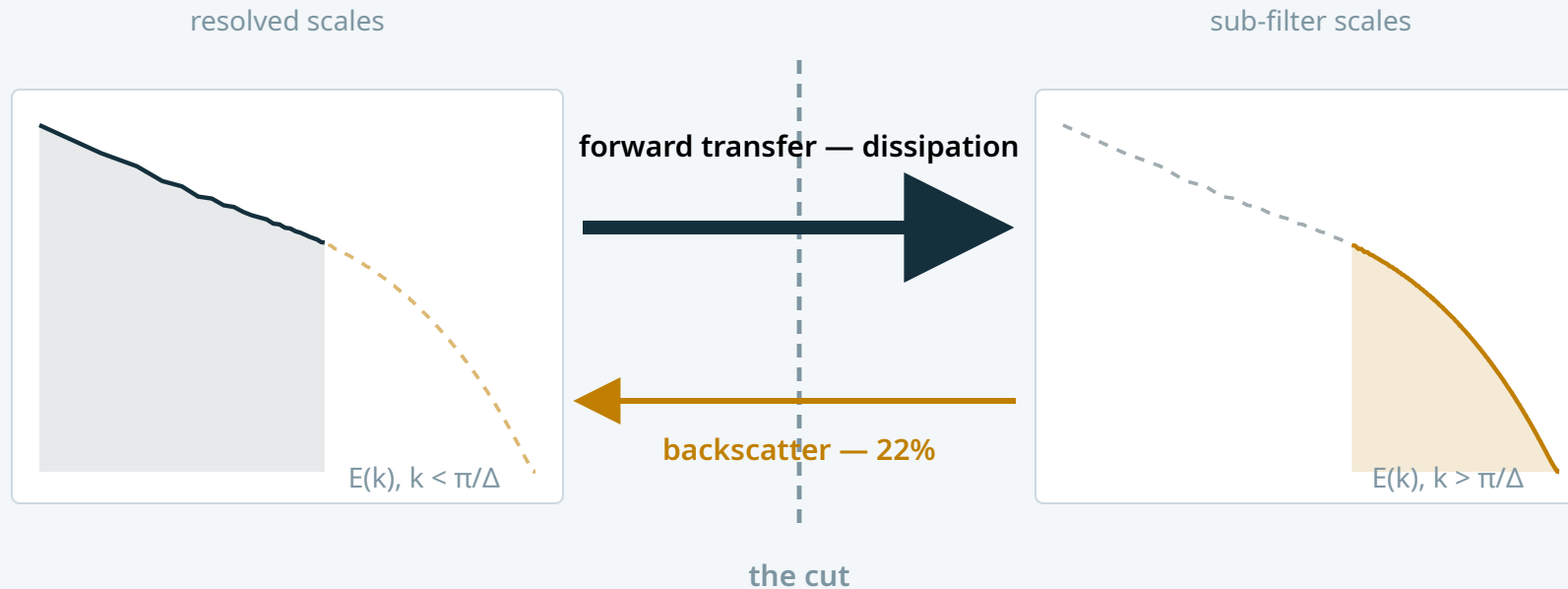
a
...rescaled – still a solution

$$x \rightarrow ax, \quad t \rightarrow bt, \quad u \rightarrow \frac{a}{b} u, \quad \nu \rightarrow \frac{a^2}{b} \nu$$

Still Navier–Stokes — but the rescaling multiplies ν by a^2/b . A closure blind to ν stays consistent only where $\mathbf{a}^2 = \mathbf{b}$, the scalings that leave ν fixed.

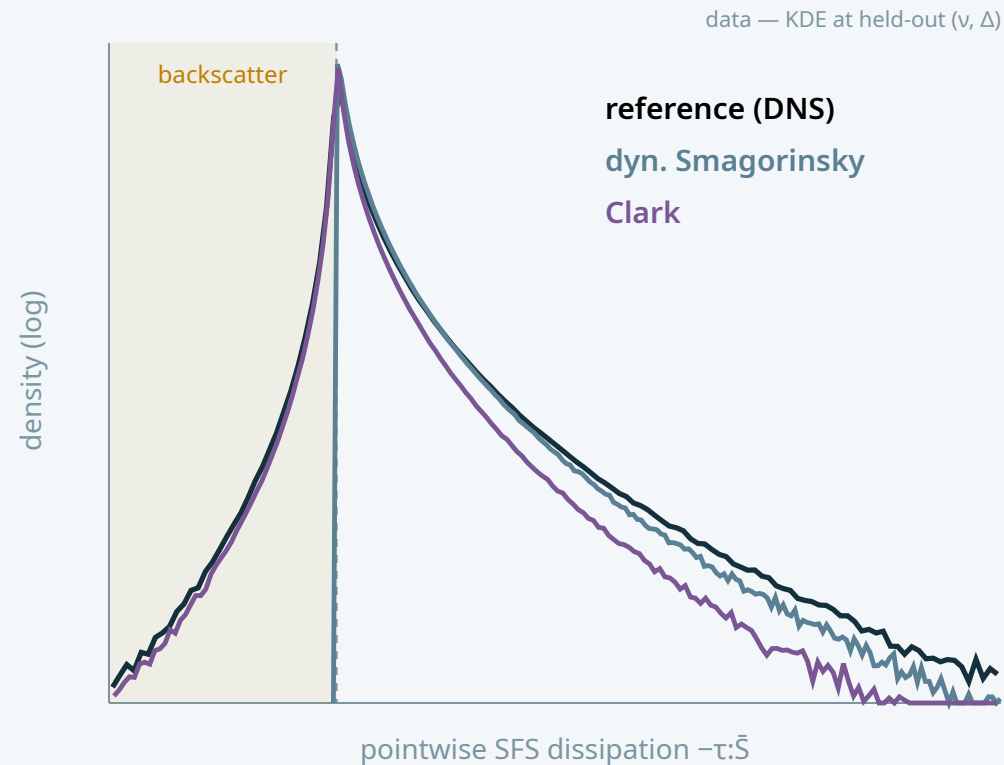
Along that family, $\bar{u}$ has a single dimensionless invariant: $\mathbf{Re}_\Delta$. Symmetry itself hands the closure its Reynolds input.

Energy crosses the cut in both directions



$\Pi = -\tau : \bar{S}$ a closure must get this net flux right — at every Reynolds number

Classical closures pick one side of this trade



FUNCTIONAL · SMAGORINSKY

$$\tau = -2 (C_s \Delta)^2 |\bar{S}| \bar{S}$$

dissipation $\approx$ right · structure wrong
backscatter 0.0004 – can't flow backward

STRUCTURAL · CLARK

$$\tau = \frac{\Delta^2}{12} \bar{A} \bar{A}^T$$

structure $\approx$ right · dissipation 0.6× too weak
backscatter 0.25 – close to the true 0.22

and notice: **v** appears in neither formula **no v**

Smagorinsky 1963 · Germano et al. 1991 · Clark et al. 1979

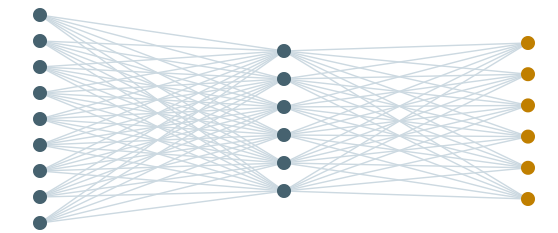
Or learn the closure from DNS — building in the physics we know

$$m : \underbrace{\bar{A}(x)}_{\mathbb{R}^{3 \times 3}} \mapsto \underbrace{\tau(x)}_{\text{sym. deviatoric}}$$

one network, fit a priori (least squares on τ) · 3 viscosities × 3 filter widths · 5 seeds

more inductive bias built in — less for the network to learn

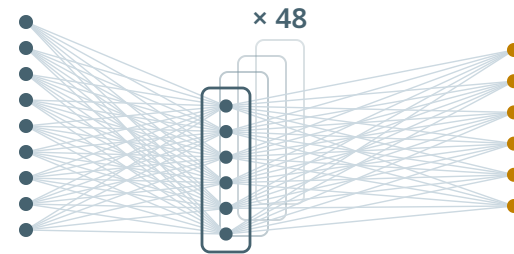
MLP · NO SYMMETRY



$\bar{A}(9)$ hidden $\tau(6)$
9 gradients in → 6 stresses out

any structure must be **learned** from data

G-CNN · TIED WEIGHTS

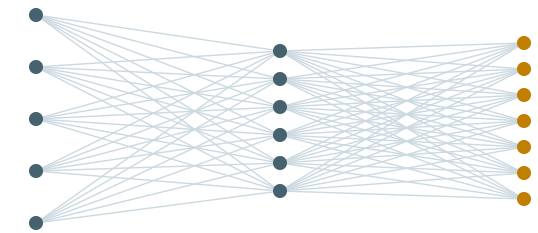


$\bar{A}(9)$ hidden $\tau(6)$
one map, tied across 48 grid symmetries

equivariant by **weight sharing**

Cohen & Welling 2016

TBNN · TENSOR BASIS



$\lambda(5)$ hidden $\alpha(7)$
 $\lambda = 5$ invariants of $\bar{A}$
 $\tau = \sum \alpha_k T_k$ · 7 fixed basis tensors

equivariant by **construction**

Ling et al. 2016

However much we build in, all three stay **single-point closures** — each reads $\tau(x)$ from $\bar{A}(x)$ alone. Hold that thought.

The learned closures are just as blind

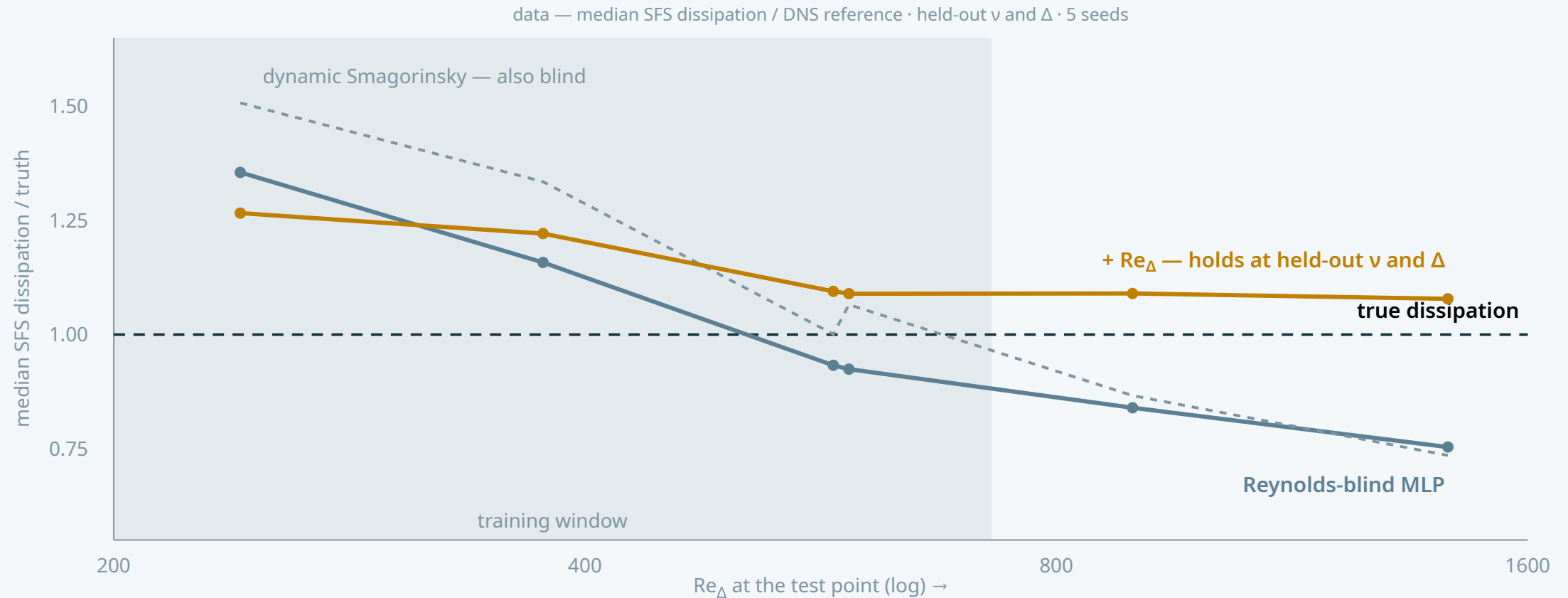
$$\tau = \Delta^2 |\bar{A}|^2 m\left(\bar{A}/|\bar{A}|\right) \text{ (no } \nu \text{)}$$

the normalization that grants scale-invariance also **erases the Reynolds number** —
the dissipation calibration freezes at the training regime

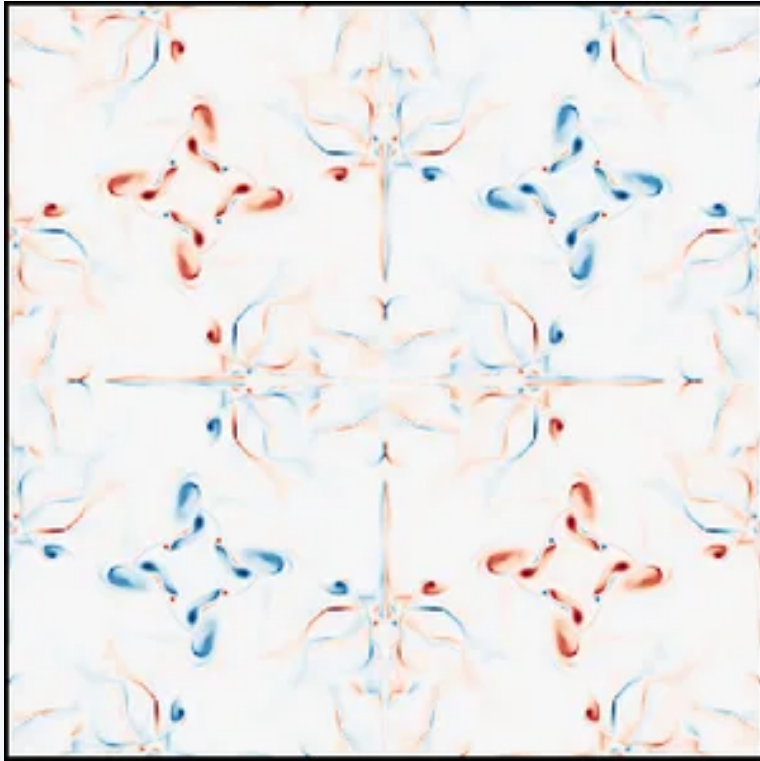
fix: train across viscosities, and give every model Re_Δ as one extra input

$$\tau = \Delta^2 |\bar{A}|^2 m\left(\bar{A}/|\bar{A}|, \text{Re}_\Delta\right) \text{ (with } \nu \text{)}$$

Tested away from training: who keeps their calibration?



A flow it has never seen: Taylor–Green becoming turbulent

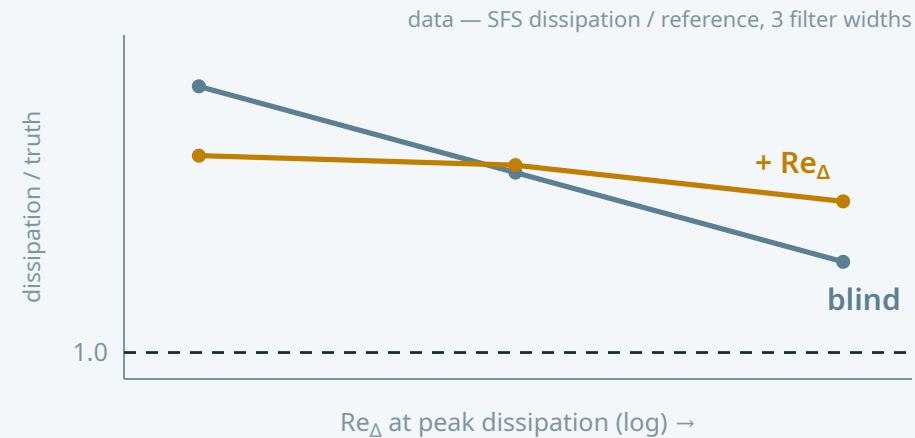


VORTICITY · $\mathbf{t} = \mathbf{0}$

decaying, transitional – out of distribution twice over

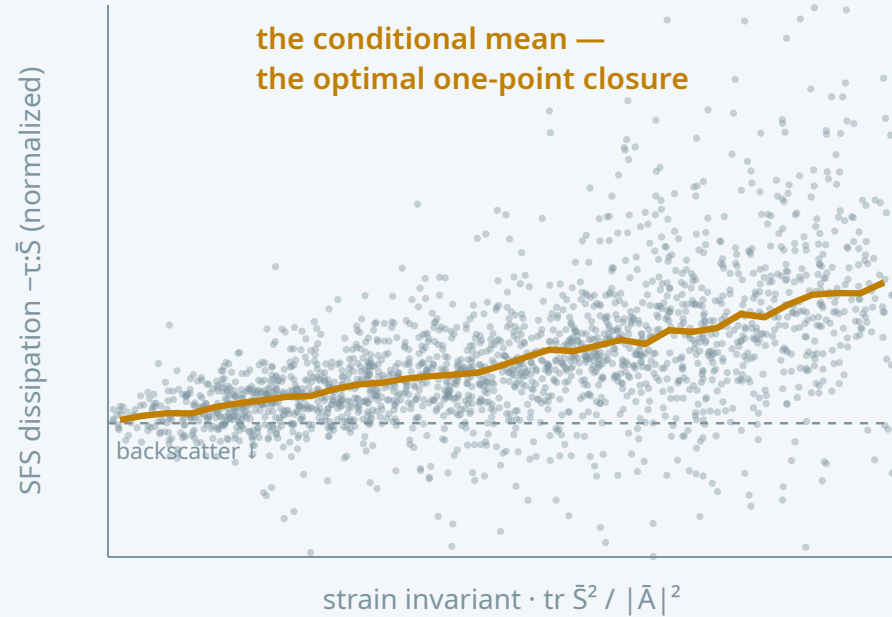
laminar → transition → decay

all closures stay stable — but over-dissipate

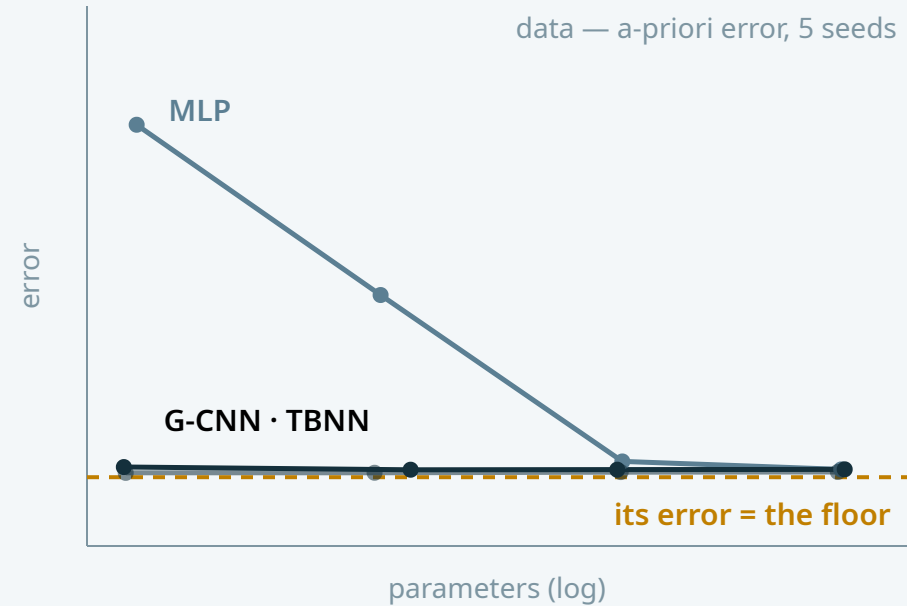


Re_Δ corrects the Reynolds-driven share of the excess,
and honestly leaves the share owed to different flow physics

Couldn't a better architecture have fixed this? We checked.



real data — one slice · one input value $\rightarrow$ a cloud of true stresses



every architecture saturates to the **same** one-point optimum — the equivariant ones with $\sim 25\times$ fewer parameters

so the Reynolds fix could never come from the architecture — **it had to come from the input**, available to all of them

TAKEAWAY

A closure that knows *what it's missing*
works where it has never been.

Agdestein & Sanderse · CWI, Amsterdam · TU Eindhoven

*"Approaching the optimal closure: equivariance, inductive bias,
and Reynolds-number generalization in data-driven LES"*

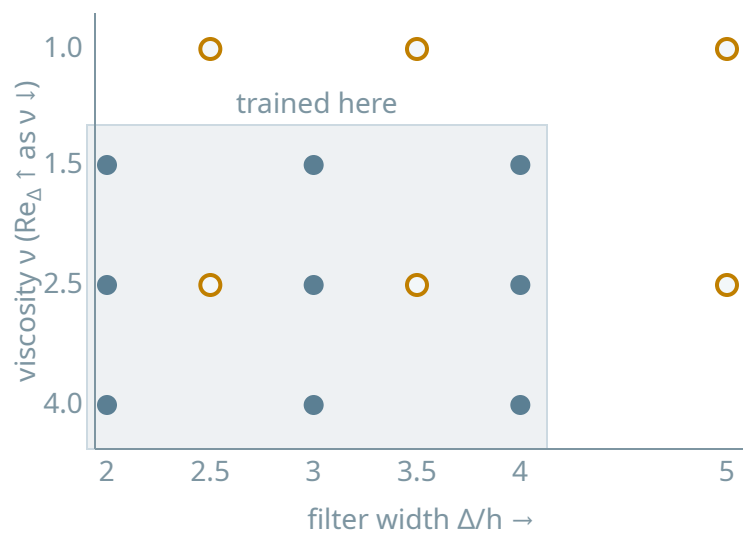
arxiv preprint: **2603.05325** · code on GitHub



Read the interactive story of this work
<https://agdestein.github.io/>

Setup: data and training

EXPERIMENT DESIGN



• trained (3×3) · ○ tested – held-out seed & ν , off-grid Δ

every test point sits off the training grid — in filter width *and* in Re_Δ

train $Re_\Delta \approx 104\text{--}727$ · 8 snapshots per grid point $\rightarrow$ 72 gradient-stress fields

DATA & TRAINING

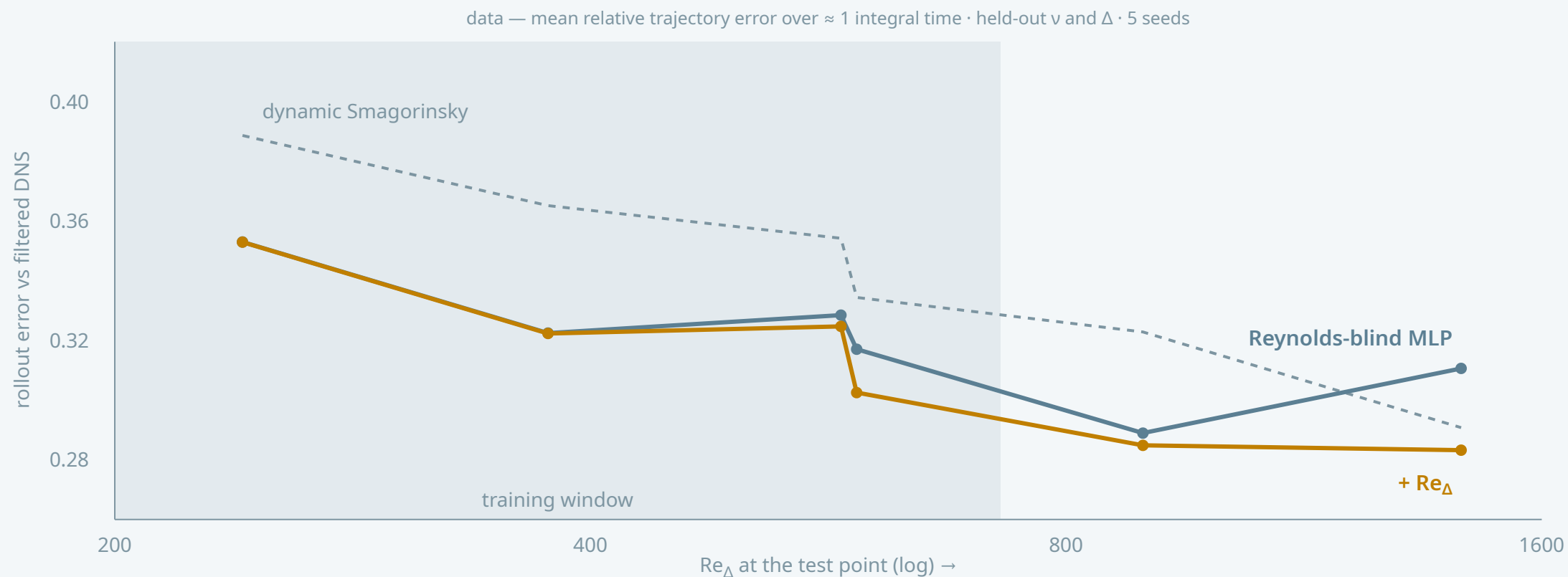
DNS — forced HIT, pseudo-spectral 810^3 · one H100
2/3-rule dealiasing · $k_{\max}\eta \geq 1.1$ · double precision

Filter — modified Gaussian onto a 128^3 LES grid
kernel commutes with the rescaling symmetry

Closures — MLP · G-CNN · TBNN, four sizes ($\approx 120\text{--}3000$ params)
5 seeds each · a-priori least squares on the normalized stress · AdamW, 20 epochs

A posteriori — rollouts vs 40 filtered-DNS snapshots (≈ 1 integral time)
error = mean relative trajectory error · baselines: no-model
· dynamic Smagorinsky · Clark

The calibration gain survives an actual LES run



inside the training window the two are essentially tied; the **Re_Δ input** pulls ahead at the highest Re (−9% at the farthest test point)

Scoreboard at the in-distribution test point ($\Delta/h = 2.5$)

CLOSURE	A-PRIORI STRESS ERROR	A-POSTERIORI ROLLOUT ERROR
No-model	1.000	0.432
Dynamic Smagorinsky	0.964	0.389
Clark	0.466	0.364
MLP	0.449	0.353
G-CNN	0.449	0.355
TBNN	0.446	0.351
conditional-mean floor	0.439	—

a priori, the data-driven closures sit on the one-point floor; a posteriori the spread narrows but the ranking holds — the learned closures stay ahead of every classical baseline

learned closures: largest tier (≈ 3000 parameters), median over 5 seeds · all metrics at held-out seed $v = 2.5 \cdot 10^{-4}$, $\Delta/h = 2.5$